

ECOM30001/90001 Basic Econometrics

Tutorial 8: Heckman's Sample Selection Model (`heckit`)

Zheng Fan

Department of Economics, The University of Melbourne

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Tutorial 8 Overview

Today's Core Goals

- **Sample Selection Bias:** When and why OLS fails on non-random samples ($lfp = 1$).
- **Heckman's 2-Step Method:** Step 1 (Probit $\rightarrow$ IMR $\hat{\lambda}_i$) $\rightarrow$ Step 2 (Augmented OLS $\rightarrow$ Test $\rho = 0$).
- **Exclusion Restrictions:** Economic rationale and econometric identification.
- **R & Exam Practice:** Manual IMR vs. `heckit()`, plus Exam MCQ.

Lecture Revision: Selection Bias & The Heckman Solution

The Problem: Incidental Truncation

Suppose we want to estimate wage offers: $\log(\text{wage}_i) = \mathbf{x}_i' \beta + \epsilon_i$.

Wages are only observed for workers ($\text{lf}_i = 1$). If unobserved factors affecting wages (ϵ_i) correlate with participation (u_i), then:

$$\text{Corr}(\epsilon_i, u_i) = \rho \neq 0 \implies E[\epsilon_i \mid \text{lf}_i = 1] \neq 0 \implies \text{OLS is biased!}$$

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The Heckman (1979) Solution: Inverse Mills Ratio (IMR)

Conditioning on working yields the augmented regression:

$$E[\log(\text{wage}_i) \mid \mathbf{w}_i, \text{lfp}_i = 1] = \mathbf{x}'_i\beta + \underbrace{\rho\sigma_\epsilon}_{\beta_\lambda} \cdot \lambda_i(l_i), \quad \text{where } \lambda_i(l_i) = \frac{\phi(l_i)}{\Phi(l_i)}$$

- $l_i = \mathbf{w}'_i\gamma$ is the probit selection index, $\phi(\cdot)$ is standard normal PDF, $\Phi(\cdot)$ is CDF.
- Including estimated $\hat{\lambda}_i$ as an extra regressor restores consistency to OLS!

Lecture Revision: Exclusion Restrictions & Workflow

Part (c)(i): Why do we need Exclusion Restrictions?

Variables included in Selection ($\mathbf{w}_i$) but **excluded from Outcome** ($\mathbf{x}_i$):

- **Economic Rationale:** Factors like nonwork income (`nwinc`) and young children (`kids1t6`) shift a woman's *reservation wage* (work decision), but do not directly dictate market *wage offers* once education and experience are held constant.
- **Econometric Reason:** Prevents severe multicollinearity between $\hat{\lambda}_i$ and $\mathbf{x}_i$.

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Heckman Two-Step Workflow:

- ① **Step 1 (Full sample, $N = 753$):** Estimate Probit model for $\text{lfpr}_i \rightarrow$ compute $\hat{\lambda}_i = \frac{\phi(\hat{l}_i)}{\Phi(\hat{l}_i)}$.
- ② **Step 2 (Workers only, $n = 428$):** Run OLS of $\log(\text{wage}_i)$ on $\mathbf{x}_i$ and $\hat{\lambda}_i$.
- ③ **Test $\rho = 0$:** t -test on $\hat{\beta}_\lambda$ ($\beta_\lambda = \rho\sigma_\epsilon = 0$).

Empirical Setting: Mroz (1987) Married Women Data

Dataset: `mroz.csv` ($N = 753$ Married Women in 1975)

- **Working Sample** ($lfp = 1$): $n_1 = 428$ women (wages observed).
- **Non-working Sample** ($lfp = 0$): $n_0 = 325$ women (wages unobserved).

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Variable Definitions:

- $lfp = 1$ if worked in 1975, 0 otherwise
- $wage$ = average hourly earnings in 1975 dollars
- $educ$ = years of schooling
- $exper$ = years of labour market experience
- $nwinc$ = nonwork income = $faminc - (wage \times hours)$
- age = woman's age in years
- $kidslt6$ = number of children aged < 6 years in household
- $kidsge6$ = number of children aged 6–18 years in household

Step 1: Probit Selection Equation for Participation

Regressor	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.2701	0.5081	0.532	0.5950
educ	0.1309***	0.0254	5.154	< 0.001
exper	0.1233***	0.0188	6.575	< 0.001
exper ²	-0.0019**	0.0006	-3.145	0.0017
nwinc	-0.000012*	0.000005	-2.434	0.0149
age	-0.0529***	0.0085	-6.246	< 0.001
kidslt6	-0.8683***	0.1184	-7.335	< 0.001
kidsge6	0.0360	0.0440	0.818	0.4135

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Key Insights from Step 1:

- **Dominant effect:** kidslt6 has a massive negative coefficient ($-0.8683, z = -7.34$). Young children are the primary barrier to labor force participation.
- **Older kids (kidsge6):** Positive ($+0.0360$) but statistically insignificant ($p = 0.41$).

Step 1 (Part 2): Computing the IMR in R

Constructing $\hat{\lambda}_i$ from Estimated Probit Index

$$\hat{l}_i = \hat{\gamma}_0 + \hat{\gamma}_1 \text{educ}_i + \cdots + \hat{\gamma}_7 \text{kidsge6}_i, \quad \hat{\lambda}_i = \frac{\phi(\hat{l}_i)}{\Phi(\hat{l}_i)}$$

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R Code Implementation:

```
# 1. Estimate probit selection model
```

```
tut8_sel <- glm(lfp ~ educ + exper + I(exper^2) + nwinc +  
               age + kidslt6 + kidsge6,  
               data = tut8_heckit, family = binomial(link="probit"))
```

```
# 2. Predict linear index I_hat
```

```
index_sel <- predict(tut8_sel)
```

```
# 3. Compute IMR = dnorm(I) / pnorm(I)
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tut8_heckit$imr1 <- dnorm(index_sel) / pnorm(index_sel)
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► **Check:** Verify with `invMillsRatio(tut8_sel)$IMR1` from `sampleSelection`.

Step 2: Augmented Outcome Equation Estimation

Part (d): OLS with Inverse Mills Ratio on Working Sample
($n = 428$)

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 + \beta_\lambda \hat{\lambda}_i + v_i$$

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Regressor	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.5781	0.3067	-1.885	0.0602
educ	0.1091***	0.0156	6.987	< 0.001
exper	0.0439**	0.0164	2.684	0.0076
exper ²	-0.0009*	0.0004	-1.946	0.0523
imr1 ($\hat{\lambda}_i$)	0.0323	0.1344	0.240	0.8104

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► **Replication Verified:** Matches `heckit(..., method="2step")` output exactly!

Comparing OLS vs. Heckit: Is Selection Bias Present?

Regressor	Unadjusted OLS	Heckit (2-Step)
(Intercept)	-0.5220	-0.5781
educ	0.1075	0.1091
exper	0.0416	0.0439
exper ²	-0.0008	-0.0009
imr1 ($\hat{\beta}_\lambda$)	—	0.0323 ($p = 0.809$)
$\hat{\sigma}_\epsilon$	0.6664	0.6636
$\hat{\rho}$	—	0.0486

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Part (e): Testing for Sample Selection Bias ($H_0 : \rho = 0$)

- Coefficient on imr1: $\hat{\beta}_\lambda = \hat{\rho}\hat{\sigma}_\epsilon = 0.0323$ ($t = 0.241$, $p = \mathbf{0.809}$).
- Since $p = 0.809 \gg 0.05$, we **fail to reject** $H_0 : \rho = 0$.
- **Conclusion:** There is **no statistical evidence of sample selection bias** in this dataset. Unadjusted OLS is reliable here.

Economic Interpretation: Why is $\rho \approx 0$?

Part (e)(iv): What does $\rho = 0$ mean in plain language?

$\rho = \text{Corr}(\epsilon_i, u_i) = 0$ means the unobserved determinants of wage offers and the unobserved determinants of employment decisions are **uncorrelated**.

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Economic Explanation:

- **Wage offer drivers:** Predominantly productivity and human capital accumulation (educ, exper).
- **Labor supply drivers:** In 1975, married women's participation decisions were largely governed by domestic constraints (presence of young children, husband's earnings).
- Because market wage determinants and home-production constraints operate through separate mechanisms, selection into work does not distort wage returns.

Exam Practice Question (MCQ)

Question: Neonatal ICU Admission Probit Model

$$\Pr(\text{ICU}_i = 1) = \Phi(-2.532 - 0.074\text{Married}_i + 0.348 \ln \text{BMI}_i)$$

Sample mean BMI = 27.66.

What is the estimated probability that an **unmarried mother** (Married = 0) with BMI = 27.66 has her baby admitted to ICU?

- ① between 1 and 5 percent
- ② between 6 and 10 percent
- ③ between 11 and 15 percent
- ④ none of the above

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Suggested Answer: Option 2 (between 6 and 10 percent)

$$\hat{l}_i = -2.532 - 0.074(0) + 0.348 \times \ln(27.66)$$

$$= -2.532 + 0.348(3.3200) = -1.3766 \approx -1.38$$

$$\hat{P}_i = \Phi(-1.38) = 1 - \Phi(1.38) = 1 - 0.9162 = \mathbf{0.0838} \text{ (8.38\%)}$$

Mid-Semester Test (MST) Paper Return

MST return

- Please collect your **MST paper** now.
- I will call your student ID (last 3 digits), please come collect your paper.
- Review the questions and clarify any concepts.
- Feel free to stay, discuss or ask any questions!